

TryHackMe — Pre-Security Learning Path

The OSI Model

Learning Notes — The 7-Layer Networking Framework

"The OSI model is the universal language of networking — without it, devices from different manufacturers could not communicate."



1. What Is the OSI Model?

The OSI Model (Open Systems Interconnection Model) is a conceptual framework developed by the International Organization for Standardization (ISO) that defines how communication should occur between devices on a computer network. It is one of the most fundamental concepts in all of networking and cyber security.

Why the OSI Model Was Created

Before the OSI model was standardised, devices from different manufacturers could not reliably communicate with each other. Each vendor had their own proprietary networking system. The OSI model solved this by creating a universal, vendor-neutral framework — a common language that all devices could follow, regardless of who made them.

1.1 Key Properties

- Consists of exactly 7 layers, numbered from Layer 1 (bottom) to Layer 7 (top).
- Each layer has a specific, well-defined set of responsibilities.
- Layers communicate only with the layers directly above and below them.
- Data travels down the stack on the sender's side and up the stack on the receiver's side.
- At each layer, information (headers/trailers) is added to the data — this process is called Encapsulation.
- The reverse process — stripping headers as data moves up the stack — is called De-encapsulation.

1.2 Mnemonic — Remembering the Layers

Layer #	Layer Name	PDU Name	Memory Aid (Top→Bottom)
7	Application	Data	All
6	Presentation	Data	People
5	Session	Data	Seem
4	Transport	Segment	To
3	Network	Packet	Need
2	Data Link	Frame	Data
1	Physical	Bits	Processing

Mnemonic

Top→Bottom: "All People Seem To Need Data Processing" | Bottom→Top: "Please Do Not Throw Sausage Pizza Away". Use whichever sticks. Knowing which layer number maps to which name is essential — you will regularly see terms like "Layer 3 switch" or "Layer 7 firewall" in real-world security work.

2. Encapsulation & De-encapsulation

Encapsulation is the process by which each layer of the OSI model wraps the data it receives from the layer above with its own header (and sometimes trailer) containing layer-specific information. The result is that by the time data reaches Layer 1, it contains the headers of all seven layers — like a set of Russian dolls.

2.1 How Encapsulation Works (Sending)

Layer	PDU Name	What Gets Added
L7	Data	Application data — the actual content (e.g. an HTTP request, email body).
L6	Data	Format/encoding information added — data translated to standard format, encryption applied.
L5	Data	Session identifiers added — establishes, manages, and synchronises the communication session.
L4	Segment	TCP/UDP header added — source/destination port numbers, sequence numbers, error checking.

L3	Packet	IP header added — source and destination IP addresses, TTL, routing information.
L2	Frame	MAC header and trailer added — source/destination MAC addresses, error detection (CRC).
L1	Bits	Converted to electrical signals, light pulses, or radio waves for physical transmission.

De-encapsulation

When data arrives at the destination device, the process reverses. Starting at Layer 1 (Physical), each layer reads and strips its own header, passes the remaining data up to the next layer, until the application at Layer 7 receives the original clean data. This is de-encapsulation.

3. The 7 Layers — Deep Dive

Layer 7 — Application | PDU: Data

The topmost layer — the one that users interact with directly. It provides networking services to applications running on a computer, acting as the interface between the network and the software the user sees.

Key Devices: End-user devices and applications (browsers, email clients) **Protocols/Standards:** HTTP, HTTPS, FTP, SMTP, DNS, POP3, IMAP, SSH

Responsibilities

- Provides network services directly to end-user applications (web browsers, email clients, file transfer programs).
- Facilitates communication between software applications and lower network layers.
- Provides a graphical user interface (GUI) through which humans interact with network data.
- Common protocols at this layer: HTTP (web browsing), FTP (file transfer), SMTP (sending email), DNS (name resolution), SSH (secure remote access).
- Every application that sends or receives data over a network uses Layer 7 services.

Security Relevance: Layer 7 firewalls (WAFs — Web Application Firewalls) inspect and filter traffic at this layer. Attacks like SQL injection, XSS, and application-level DDoS all target Layer 7. HTTPS encryption originates here.

Layer 6 — Presentation | PDU: Data

The translator of the OSI model. It ensures that data sent from one application can be read and understood by an application on a different system, even if the two systems use different data formats or encodings.

Key Devices: Operating system components, TLS/SSL libraries **Protocols/Standards:** TLS/SSL, JPEG, PNG, GIF, ASCII, EBCDIC, MPEG, MIME

Responsibilities

- Translates data between the format the application understands and a standardised network format.
- Handles data encoding and decoding — e.g. converting text between ASCII and EBCDIC character sets.
- Manages data compression to reduce the amount of data that needs to be transmitted.
- Handles encryption and decryption — HTTPS security (TLS) operates at this layer.
- Ensures that data sent from a Windows machine can be understood by a Linux machine, for example.

Security Relevance: TLS/SSL encryption and decryption happens at Layer 6 — this is the foundation of HTTPS. A man-in-the-middle (MITM) attack that intercepts unencrypted traffic is exploiting the absence of Layer 6 protection.

Layer 5 — Session | PDU: Data

The session manager. It creates, maintains, synchronises, and terminates communication sessions between applications on different hosts. A session is a semi-permanent interactive exchange of information between two devices.

Key Devices: Operating system, application layer software **Protocols/Standards:** NetBIOS, NFS, RPC, SMB, SQL sessions, SIP

Responsibilities

- Establishes a session (connection) between two communicating applications before data transfer begins.
- Maintains the session throughout the data transfer — keeping the connection alive.
- Synchronises data transfer using checkpoints — if a transfer fails, only the data after the last checkpoint needs to be resent.
- Terminates the session cleanly once data transfer is complete.
- Sessions are unique — data cannot be transmitted across multiple sessions simultaneously; it is confined to its established session.
- Divides data into smaller chunks (packets) ready for the Transport layer to handle.

Security Relevance: Session hijacking attacks target Layer 5 — an attacker steals a valid session token to take over an authenticated session without needing credentials. Session fixation is another Layer 5 attack vector.

Layer 4 — Transport | PDU: Segment / Datagram

The reliability and delivery control layer. It governs how much data is sent, at what rate, and to which destination port. It selects the appropriate protocol — TCP for reliability or UDP for speed — depending on the application's requirements.

Key Devices: Endpoints (sender and receiver software) **Protocols/Standards:** TCP (Transmission Control Protocol), UDP (User Datagram Protocol)

Responsibilities

- Chooses the transmission protocol: TCP (reliable, connection-oriented) or UDP (fast, connectionless).
- TCP — guarantees delivery with error checking, acknowledgements, and retransmission of lost packets.
- UDP — sends data without confirmation, prioritising speed over reliability (used for streaming, gaming, DNS).
- Handles segmentation — breaking large data into manageable segments for transmission.
- Handles reassembly — putting segments back together in the correct order at the destination.
- Manages flow control (preventing sender from overwhelming receiver) and congestion control.
- Port numbers at this layer identify the specific application or service (e.g. port 80 = HTTP, port 443 = HTTPS, port 22 = SSH).

Security Relevance: Port scanning (Nmap) operates at Layer 4 — discovering which TCP/UDP ports are open on a target. Firewall rules commonly filter at Layer 4. SYN flood attacks (TCP) and UDP flood attacks target this layer.

Layer 3 — Network | PDU: Packet

The routing and addressing layer. It is responsible for determining the optimal path for data to travel from source to destination across multiple networks, using logical addressing (IP addresses).

Key Devices: Routers, Layer 3 switches **Protocols/Standards:** IPv4, IPv6, OSPF, RIP, ICMP, BGP

Responsibilities

- Assigns and uses logical IP addresses (IPv4 and IPv6) to identify source and destination devices.
- Determines the best route (path) for packets to travel from sender to receiver across networks.
- Uses routing protocols (OSPF, RIP, BGP) to dynamically discover and update routing tables.
- Handles packet fragmentation and reassembly when packets are too large for a network segment.
- Operates through routers — devices that read IP headers and forward packets toward their destination.
- Routing decisions consider: path length (fewest hops), reliability, and bandwidth of available routes.

Security Relevance: IP spoofing, ICMP-based attacks (ping floods, Smurf attacks), and routing protocol manipulation (BGP hijacking) all target Layer 3. Traceroute is a Layer 3 diagnostic tool. Network-layer ACLs and routing security are defences.

Layer 2 — Data Link | PDU: Frame

The physical addressing layer. It handles delivery of data within a single network segment (LAN), using MAC addresses to identify the exact source and destination device on that local network. It also ensures data is in a suitable format for transmission.

Key Devices: Switches, Network Interface Cards (NICs), bridges **Protocols/Standards:** Ethernet, Wi-Fi (802.11), ARP, PPP, MAC addressing

Responsibilities

- Adds MAC (Media Access Control) addresses to frames — the physical hardware address of the NIC.
- Every NIC has a unique MAC address assigned by the manufacturer and burned into the hardware.
- Handles error detection using CRC (Cyclic Redundancy Check) in the frame trailer.
- Controls access to the physical medium — determining when a device is allowed to transmit.
- Presents data in a format suitable for transmission over the physical medium (Layer 1).
- Operates through network switches — devices that read MAC addresses and forward frames within a LAN.
- ARP (Address Resolution Protocol) operates between Layer 2 and Layer 3 — resolving IP addresses to MAC addresses.

Security Relevance: MAC address spoofing, ARP poisoning/spoofing, and VLAN hopping are all Layer 2 attacks. Switches can be hardened with port security, dynamic ARP inspection, and DHCP snooping to defend at this layer.

Layer 1 — Physical | PDU: Bits

The hardware layer. It defines how raw binary data (0s and 1s) is transmitted as physical signals over the network medium — whether that is electrical signals over copper cables, light pulses over fibre optic cables, or radio waves for wireless.

Key Devices: Cables (Ethernet/fibre), hubs, repeaters, network cards, Wi-Fi antennas

Protocols/Standards: Ethernet (802.3), USB, Bluetooth, Wi-Fi (802.11 physical), DSL

Responsibilities

- Transmits and receives raw binary data (bits) as physical signals.
- Defines the physical medium: copper cables (Ethernet), fibre optic cables, or wireless radio waves.
- Defines electrical, optical, and mechanical specifications — voltage levels, cable types, connector shapes.
- Ethernet cables are the standard wired physical medium for LANs.
- Hubs operate at Layer 1 — they repeat all incoming signals to all ports without any intelligence.
- Wi-Fi transmits data as radio waves — the physical layer equivalent of a wireless network.
- Data at this layer is simply a stream of bits — no addressing, no error correction, just signal transmission.

Security Relevance: Physical layer attacks include: wiretapping (passive eavesdropping on cables), signal jamming (disrupting wireless), and hardware keyloggers. Physical security is the last line of defence — if an attacker has physical access, all higher-layer controls can be bypassed.

4. TCP vs. UDP — The Transport Layer Protocols

The Transport layer (Layer 4) uses two primary protocols. Understanding their difference is fundamental to networking and security work.

Feature	TCP — Transmission Control Protocol	UDP — User Datagram Protocol
Connection Type	Connection-oriented — establishes a connection first (3-way handshake)	Connectionless — sends data immediately without setup
Reliability	Guaranteed delivery — lost packets are retransmitted	No guaranteed delivery — packets may be lost silently
Order	Data is reassembled in the correct sequence	No guarantee of order — packets may arrive out of sequence
Speed	Slower — overhead from acknowledgements and error checking	Faster — minimal overhead, no acknowledgement required
Error Checking	Full error checking with acknowledgements	Basic checksum only — no retransmission
Use Cases	Web (HTTP/HTTPS), Email (SMTP), File Transfer (FTP), SSH	Streaming, gaming, VoIP, DNS queries, DHCP
Header Size	20–60 bytes (larger)	8 bytes (much smaller)
Security Risk	SYN flood (half-open connections exhaust resources)	UDP flood (high-volume traffic overwhelms target)

The TCP 3-Way Handshake

Before TCP sends any data, it establishes a connection using three messages: (1) SYN — the client sends a synchronise request. (2) SYN-ACK — the server acknowledges and sends its own synchronise. (3) ACK — the client acknowledges. Only then does data transfer begin. A SYN flood attack sends thousands of SYN requests without completing the handshake, exhausting the server's connection table.

5. OSI vs. TCP/IP Model

The TCP/IP model is the practical, real-world implementation that the Internet actually uses. The OSI model is the theoretical reference framework. Understanding both and how they map to each other is essential.

OSI Layer #	OSI Layer Name	TCP/IP Layer	Example Protocols
7	Application	Application	HTTP, HTTPS, FTP, SMTP, DNS, SSH
6	Presentation	Application	TLS/SSL, MIME, JPEG, ASCII
5	Session	Application	NetBIOS, NFS, RPC, SIP
4	Transport	Transport	TCP, UDP
3	Network	Internet	IPv4, IPv6, ICMP, OSPF, BGP
2	Data Link	Network Interface	Ethernet, ARP, Wi-Fi (MAC layer)
1	Physical	Network Interface	Ethernet (physical), Wi-Fi (radio)

Key Difference

The TCP/IP model collapses OSI Layers 5, 6, and 7 into a single Application layer, and combines OSI Layers 1 and 2 into a single Network Interface layer. The OSI model is used for teaching and troubleshooting; TCP/IP is what actually runs the Internet.

6. Practical — The OSI Dungeon Game

The TryHackMe room includes an interactive challenge called the OSI Dungeon. A character must navigate through a dungeon by selecting the correct OSI layer doors in the right order (Layer 1 through Layer 7). Selecting the wrong door ends the game and requires restarting.

6.1 The Correct Order to Escape

Navigate through the dungeon doors in ascending layer order — from the Physical layer at the bottom to the Application layer at the top:

Door	Layer	Remember
1	Physical	The foundation — bits and cables. Start here.
2	Data Link	MAC addresses and frames — local delivery.
3	Network	IP addresses and routing — cross-network delivery.
4	Transport	TCP/UDP — ports, reliability, and segmentation.
5	Session	Connection setup and management.
6	Presentation	Translation, encryption, and formatting.
7	Application	The top — what the user interacts with. Escape here!

Flag Captured

THM{OSI_DUNGEON_ESCAPED} — Completing the dungeon in the correct order (L1→L2→L3→L4→L5→L6→L7) demonstrates that you have internalised the layer sequence. The staff record to beat is 19 seconds!

7. OSI Model in Security — Layer-by-Layer Attack Surface

Every layer of the OSI model has specific attack vectors and defensive controls. Understanding which layer an attack targets is fundamental to both offensive security (knowing where to strike) and defensive security (knowing where to protect).

Layer	Name	Common Attacks	Key Defences
L7	Application	SQL Injection, XSS, CSRF, App-DDoS, Brute Force	WAF, Input validation, MFA, Rate limiting
L6	Presentation	MITM (unencrypted), SSL stripping, Malformed data	TLS/SSL, Certificate pinning, Enforce HTTPS
L5	Session	Session hijacking, Session fixation, Cookie theft	Secure cookies, Short session timeouts, HTTPS
L4	Transport	SYN flood, Port scanning, UDP flood, TCP hijacking	Firewalls, Rate limiting, SYN cookies, Port filtering
L3	Network	IP spoofing, ICMP flood, Route hijacking (BGP)	ACLs, Ingress filtering, IPSec, IDS/IPS
L2	Data Link	ARP poisoning, MAC spoofing, VLAN hopping	Port security, Dynamic ARP Inspection, 802.1X
L1	Physical	Wiretapping, Cable cutting, Hardware keyloggers	Physical access controls, Cable management, CCTV

8. Key Takeaways

The OSI Model Is the Foundation of Everything

Every concept in networking — from packet analysis in Wireshark to configuring firewall rules — maps back to a specific OSI layer. Learning the model deeply, not just memorising the names, is a career-long investment.

Encapsulation Is How Networks Actually Work

Understanding that data gets wrapped in headers at each layer on the way down, and unwrapped on the way up, explains why protocols like Ethernet, IP, TCP, and HTTP stack on top of each other. Packet capture tools like Wireshark show you exactly these layers in action.

Layer Numbers Are a Universal Language

When a vendor says "Layer 7 firewall" or "Layer 3 switch," they mean a device that inspects traffic up to that OSI layer. A Layer 3 switch routes by IP address; a Layer 7 firewall inspects the actual application content. Knowing the layers by number is non-negotiable.

TCP and UDP Serve Different Purposes

The choice between TCP and UDP is a fundamental engineering decision. Reliability requires TCP; speed requires UDP. As a security professional, you must know which protocols use which transport, because firewall rules and attack methods depend on it.

Every Layer Is an Attack Surface

Security professionals don't just defend the application — they defend all seven layers. Physical access controls, network segmentation, firewall rules, TLS certificates, and application-layer input validation are all defences at different layers of the same model.

8.1 Next Steps on TryHackMe

- **Immediate:** Continue with the Packets and Frames room — deep-dive into Layer 2/3 data encapsulation.
- **Foundation:** Study the Extending Your Network room — VLANs, subnetting, and network design.
- **Foundation:** Complete the Intro to LAN room if you haven't — LANs, topologies, ARP.
- **Applied:** Take the Wireshark: The Basics room — see OSI layers in live packet captures.
- **Security:** Learn the Nmap room — port scanning targets Layers 3 and 4 directly.
- **Security Path:** Work through the Network Security module to see how each layer is exploited.

The OSI Model in Practice

Every time you run a web browser, all seven OSI layers activate simultaneously: L7 HTTP request → L6 TLS encryption → L5 session established → L4 TCP segments → L3 IP routing → L2 Ethernet framing → L1 electrical signals on the cable. The model is not abstract theory — it is the reality of every single network communication you make.

Source: TryHackMe — OSI Model Room | tryhackme.com/room/osimodelzi